MTH636 – Game Theory

Lecture 31: Strategies, Utility & Bayesian Nash Equilibrium

1 Extensive Form Representation of Harsanyi Games

Recall from Lecture 30 that every Harsanyi game of incomplete information can be written as an extensive form game with imperfect information and chance nodes.

Theorem: Harsanyi's Representation Theorem (restated)

Every Harsanyi game of incomplete information can be written as an **extensive form game** with imperfect information and chance nodes.

For the extensive form representation:

- The number of **information sets** for player i is $|T_i|$ – one information set per possible type of player i.
- The game will have **perfect recall** for all players. This means we may use **behavior strategies**.

2 Types of Strategies

Since the extensive form game has perfect recall, three kinds of strategies are relevant:

2.1 Pure Strategies

Definition: Pure Strategy in a Harsanyi Game

A **pure strategy** for player i is a function

$$s_i : T_i \longrightarrow \bigcup_{t_i \in T_i} A_i(t_i),$$

where $s_i(t_i) \in A_i(t_i)$ for every $t_i \in T_i$.

2.2 Mixed Strategies

Definition: Mixed Strategy in a Harsanyi Game

A **mixed strategy** for player i is a probability distribution over the pure strategies of player i.

2.3 Behavior Strategies

Definition: Behavior Strategy in a Harsanyi Game

A **behavior strategy** for player i is a function

$$b_i : T_i \longrightarrow \bigcup_{t_i \in T_i} \Delta A_i(t_i),$$

where $b_i(t_i) \in \Delta A_i(t_i)$ for every $t_i \in T_i$.
Here $\Delta A_i(t_i)$ denotes the set of probability distributions (mixed actions) over $A_i(t_i)$.

Note

In the extensive form, a behavior strategy for player i assigns, to each information set (i.e. each type $t_i \in T_i$), a probability distribution over the available actions at that information set:

$$\tilde{s}_i : H_i \longrightarrow \bigcup_{h \in H_i} A(h), \quad \tilde{s}_i(h_i) \in A(h_i).$$

This corresponds exactly to $b_i(t_i) \in \Delta A_i(t_i)$ in the type-space notation.

3 Utility in Harsanyi Games

3.1 Utility at a Type Profile (Ex Ante)

Consider a behavior strategy profile $b = (b_1, \ldots, b_n)$.

For a type profile $t = (t_1, \ldots, t_n)$ and an action profile $a = (a_1, \ldots, a_n)$ with $a_i \in A_i(t_i)$ for all $i \in \{1, \ldots, n\}$, the **utility of player i at type profile t** is:

$$V_i(t; b) = \sum_{a \in \times_{i \in N} A_i(t_i)} b_1(t_1; a_1) \times b_2(t_2; a_2) \times \cdots \times b_n(t_n; a_n) \cdot u_i(t, a),$$

where $b_j(t_j; a_j)$ denotes the probability assigned by player j's behavior strategy to action a_j when her type is t_j.

Note

This is the **ex-ante utility** at type profile t – the expected utility before the randomisation (i.e. before the mixed actions are realised), but after the type profile is fixed. It is called the utility at the *ex-ante stage*, before the randomisation takes place.

3.2 Total (Ex Ante) Utility

The **total (ex-ante) utility** of player i under behavior strategy profile b is:

$$V_i(b) = \sum_{t \in T} p(t)\, V_i(t; b).$$

3.3 Interim Utility

After player i knows her own type t_i (but before learning the types of others), she can compute her **interim (conditional) utility**:

$$U_i(b \mid t_i) = \sum_{t_{-i} \in T_{-i}} p(t_{-i} \mid t_i)\, U_i((t_i, t_{-i}),\, b)\,.$$

This is called player i's **interim utility** – she knows her own type t_i and takes expectation over the opponents' types using the conditional belief $p(t_{-i} \mid t_i)$.

3.4 Relationship Between Ex-Ante and Interim Utility

The following chain of equalities shows that the ex-ante utility can be decomposed through the interim utility:

$$\begin{aligned}
V_i(b) &= \sum_{t \in T} p(t)\, V_i(t; b) \\
&= \sum_{t \in T} p(t_{-i} \mid t_i)\, p(t_i)\, V_i(t; b) \\
&= \sum_{t_i \in T_i} \sum_{t_{-i} \in T_{-i}} p(t_i) \Big(p(t_{-i} \mid t_i)\, V_i(t; b) \Big) \\
&= \sum_{t_i \in T_i} p(t_i) \sum_{t_{-i} \in T_{-i}} p(t_{-i} \mid t_i)\, V_i(t; b) \\
&= \sum_{t_i \in T_i} p(t_i)\, U(b \mid t_i)\,.
\end{aligned}$$

Key Takeaway

The ex-ante utility is a weighted sum of interim utilities:

$$V_i(b) = \sum_{t_i \in T_i} p(t_i)\, U_i(b \mid t_i).$$

This is analogous to the law of total expectation.

4 Nash Equilibrium and Bayesian Nash Equilibrium

4.1 Nash Equilibrium (NE)

Definition: Nash Equilibrium in a Harsanyi Game

A strategy profile $\sigma^* = (\sigma_i)_{i \in N}$ is a **Nash Equilibrium (NE)** if

$$V_i(\sigma^*) \geq V_i(\sigma_i,\, \sigma^*_{-i}) \quad \forall\, i \in N \text{ and all } \sigma_i \in B_i,$$

where B_i denotes the set of all behavior strategies for player i.

4.2 Bayesian Nash Equilibrium (BNE)

The Nash Equilibrium above is the *ex-ante* equilibrium concept. A more refined concept – the **Bayesian Nash Equilibrium** – requires optimality *at the interim stage*, i.e., for every type the player might be.

Definition: Bayesian Nash Equilibrium (BNE)

A strategy profile σ^* is a **Bayesian Nash Equilibrium (BNE)** if

$$U_i(\sigma^* \mid t_i) \geq U_i\Big((a_i, \sigma^*_{-i}) \mid t_i\Big) \quad \forall\, i \in N,\ \forall\, t_i \in T_i,\ \forall\, a_i \in A_i(t_i).$$

That is, for every player i, for every type t_i, and for every possible deviation a_i, the equilibrium strategy σ_i^* is at least as good as any deviation – when evaluated using the interim (conditional) utility given type t_i.

Note

In the case where T_i has multiple elements, say t_1, t_n, t_2 (as in an example with three types), the total utility decomposes as:

$$U_i(\sigma) = p(t_1)\, U_i(\sigma \mid t_1) + p(t_n)\, U_i(\sigma \mid t_n) + p(t_2)\, U_i(\sigma \mid t_2).$$

5 Key Results

Proposition: Every BNE is also an NE

Every Bayesian Nash Equilibrium is also a Nash Equilibrium.

Proof. See Solan's book. □

Theorem: Existence of NE in Behavior Strategies

For an incomplete information game (in the Harsanyi sense), a Nash Equilibrium exists in **behavior strategies**.

Proof. The corresponding extensive form game has a Nash Equilibrium in mixed strategies. Since the game has perfect recall, every mixed strategy is equivalent to a behavior strategy. Therefore a behavior strategy Nash Equilibrium exists. □

Intuition

The key steps in the existence proof are:

1. The Harsanyi game ⇒ extensive form with chance nodes and information sets.
2. The extensive form game has a Nash Equilibrium in mixed strategies (by Nash's theorem, since the game is finite).
3. Perfect recall ⇒ every mixed strategy is equivalent to a behavior strategy

(Kuhn's theorem).
4. Therefore a *behavior strategy* NE exists.